Vanchor Helm Board — sheet index

1 carrier: 12V in, protection, 5V buck header, Orange Pi Zero 3 ribbon header, UART/I2C JSTs

2 control: Pico 2, external thrust-driver interface (J13), BTN8982 servo bridge, AS5600

carrier

control

12V battery only; 5V from generic buck module on J14
Orange Pi Zero 3 (IDC ribbon) + Pico 2; external IBT-2 thrust driver; on-board servo bridge
vanchor-ng

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File: vanchor-helm.kicad_sch

Title: Vanchor Helm Board

Size: A3	Date: 2026-07-02	**Rev: v4.1**
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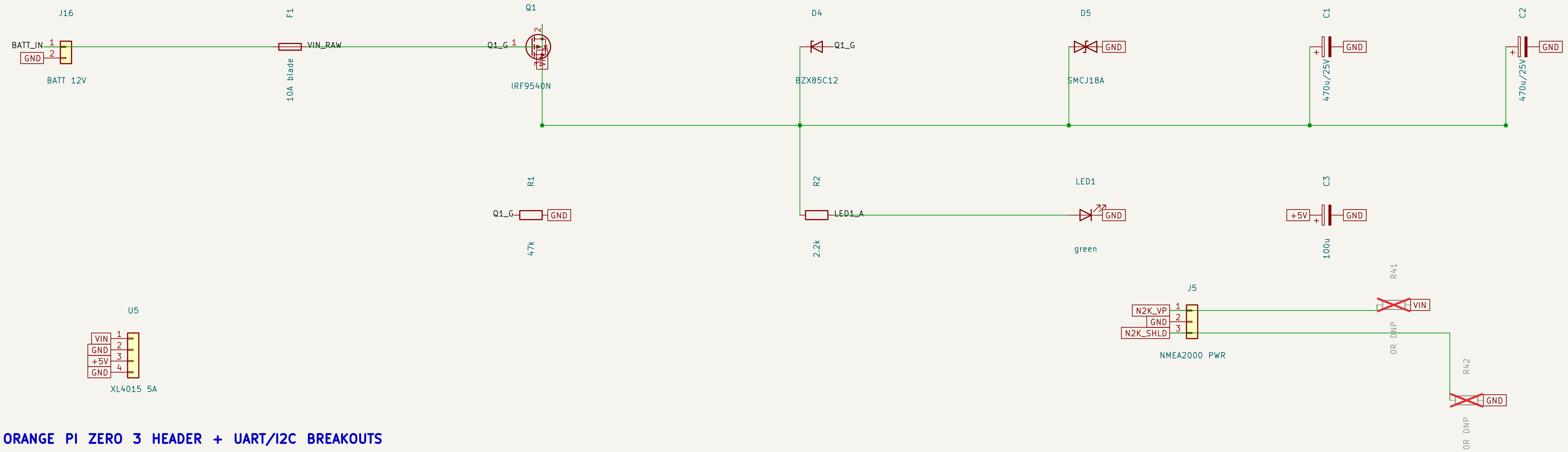
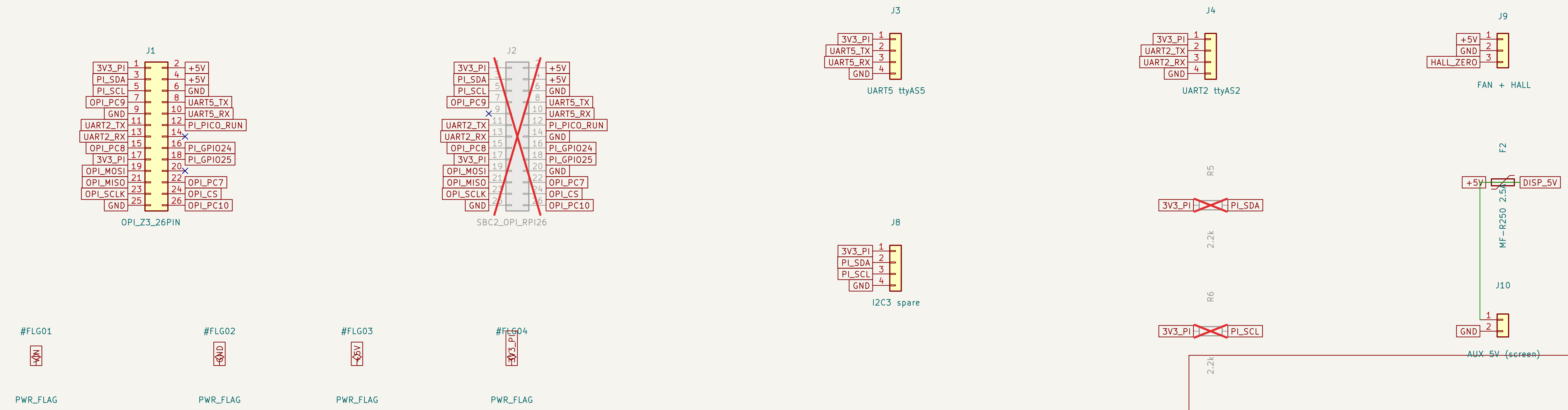
POWER: 12V battery -> F1 -> reverse-FET -> VIN; XL4015 buck daughterboard U5 (set 5.1V first)

CARRIER: J1 -> 26-way IDC ribbon -> Orange Pi Zero 3 (powered through the ribbon, no USB-C)



ORANGE PI ZERO 3 HEADER + UART/I2C BREAKOUTS



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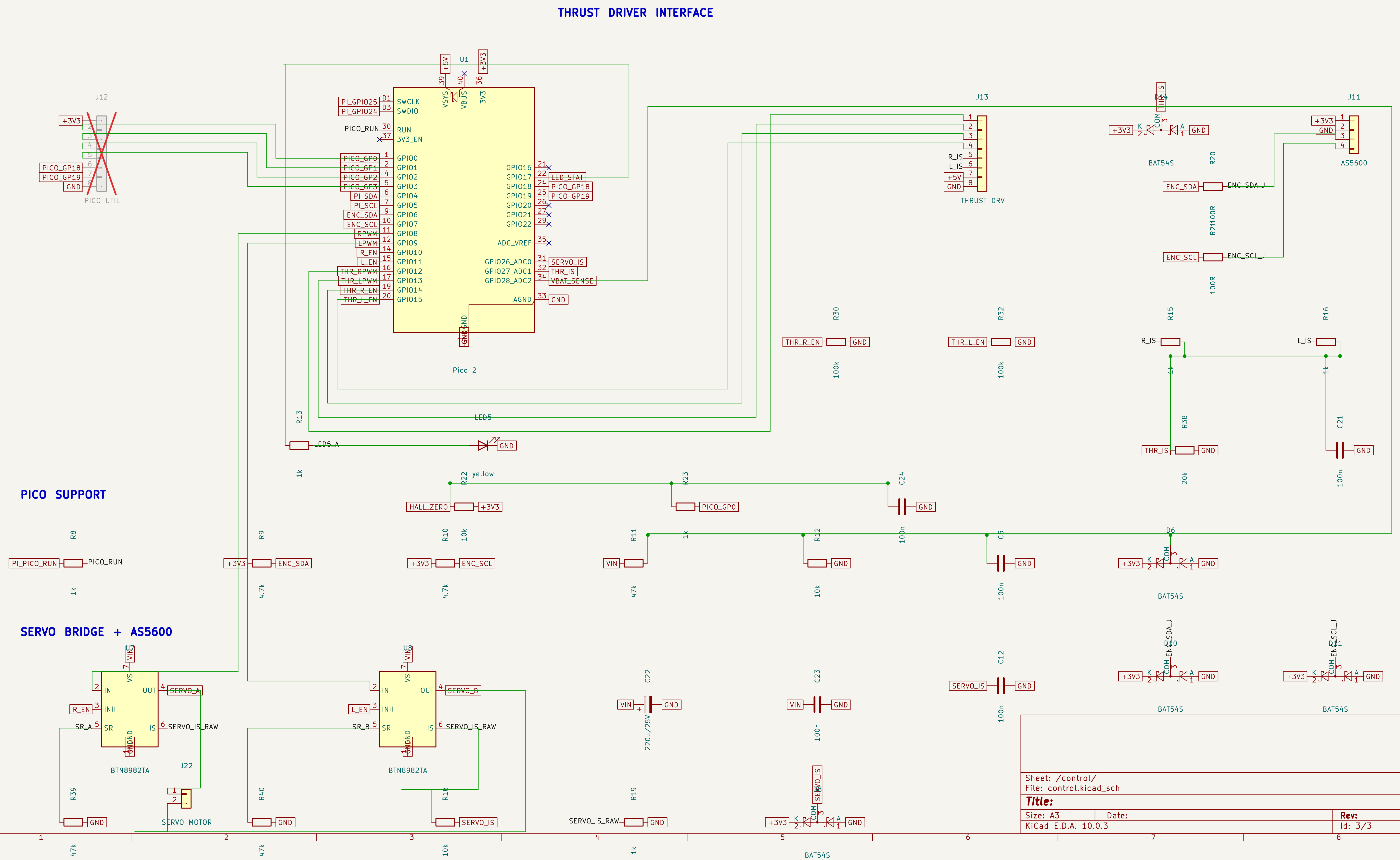
PICO 2: I2C slave 0x42 to the Zero 3; SWD reflash via ribbon pins 16/18, RUN on 12

THRUST: J13 = IBT-2 pin order to the external driver; EN pull-downs = failsafe off

SERVO: 2x BTN8982TA on protected VIN; AS5600 encoder cable on J11



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